ASSESSMENT TOOL-2

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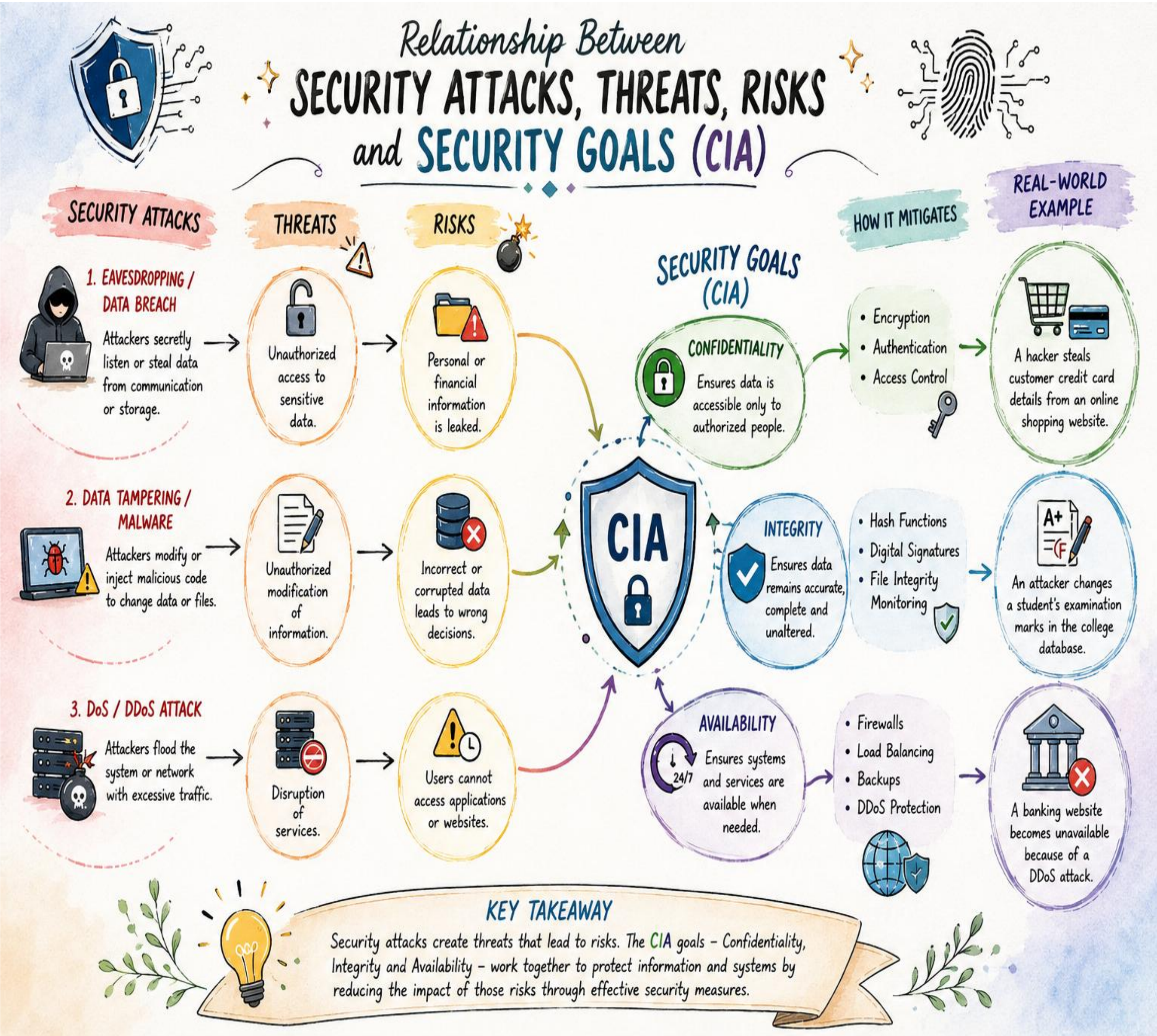
Course: Cryptography and Network Security

Course Code: CSA5113

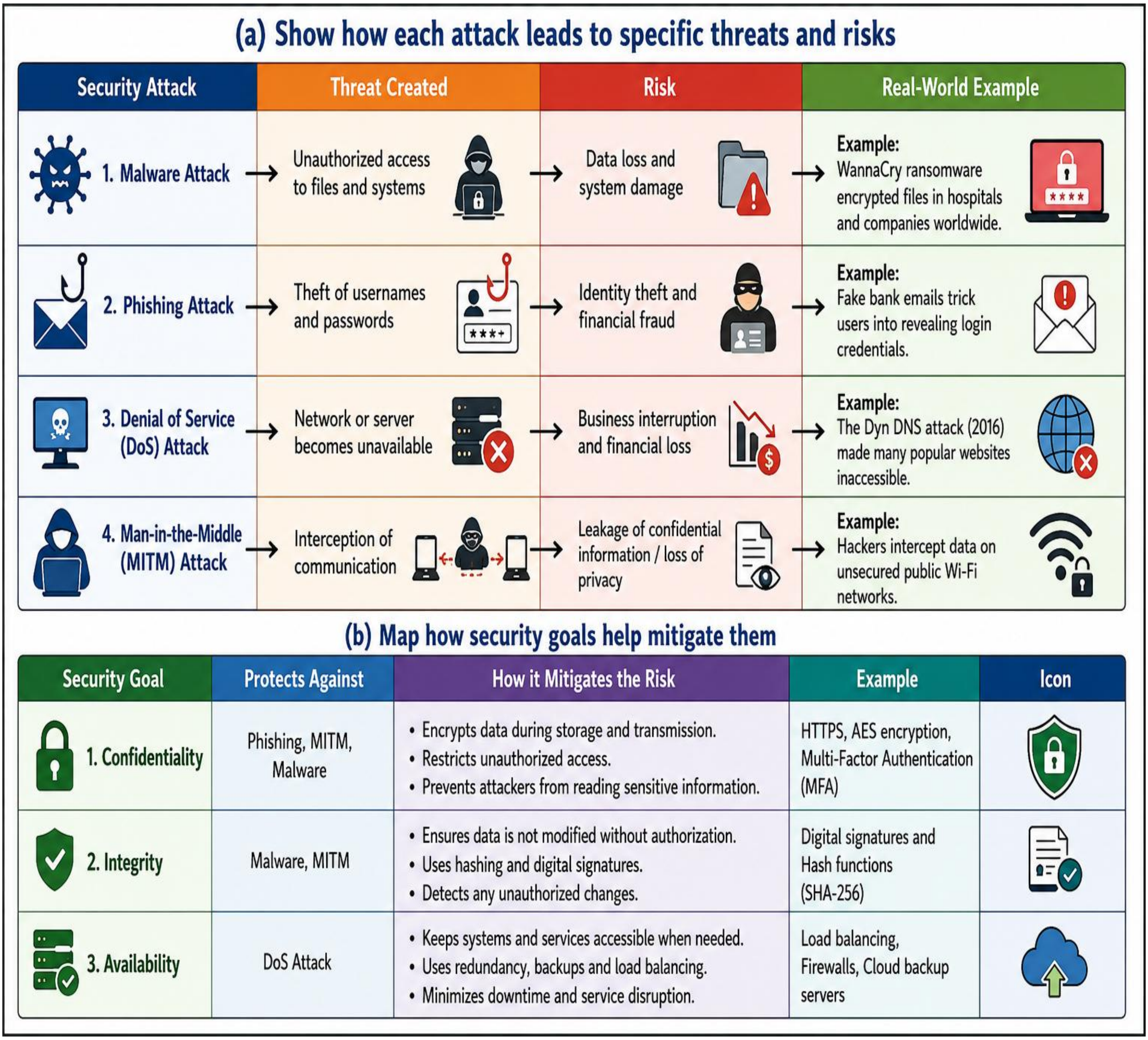
1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability. Perform the following tasks:
a. Show how each attack leads to specific threats and risks
b. Map how security goals help mitigate them Provide one real-world example for each link.

Introduction:

- Information security protects computer systems and data from unauthorized access, modification, and destruction.
- Security attacks exploit vulnerabilities in a system, creating threats that lead to various security risks.
- Organizations use the CIA Triad—**Confidentiality, Integrity, and Availability**—to protect information assets.
- Understanding the relationship between attacks, threats, risks, and security goals helps in designing secure systems.



- **Malware Attack** creates the threat of unauthorized access, leading to data loss and system damage. **Integrity** reduces this risk by using hash functions, digital signatures, and antivirus software. *(Example: WannaCry ransomware)*
- **Phishing Attack** steals usernames and passwords, resulting in identity theft and financial fraud. **Confidentiality** protects sensitive information using encryption and Multi-Factor Authentication (MFA). *(Example: Fake bank emails)*
- **Denial of Service (DoS) Attack** makes servers or networks unavailable, causing business interruption and financial loss. **Availability** ensures continuous service using firewalls, load balancing, and backup servers. *(Example: Dyn DNS Attack – 2016)*
- **Man-in-the-Middle (MITM) Attack** intercepts communication, leading to leakage of confidential information and privacy loss. **Confidentiality** and **Integrity** prevent this through encryption and secure communication protocols. *(Example: Public Wi-Fi interception)*
- The **CIA Triad (Confidentiality, Integrity, and Availability)** provides the foundation for protecting information systems against cyber threats and minimizing security risks.



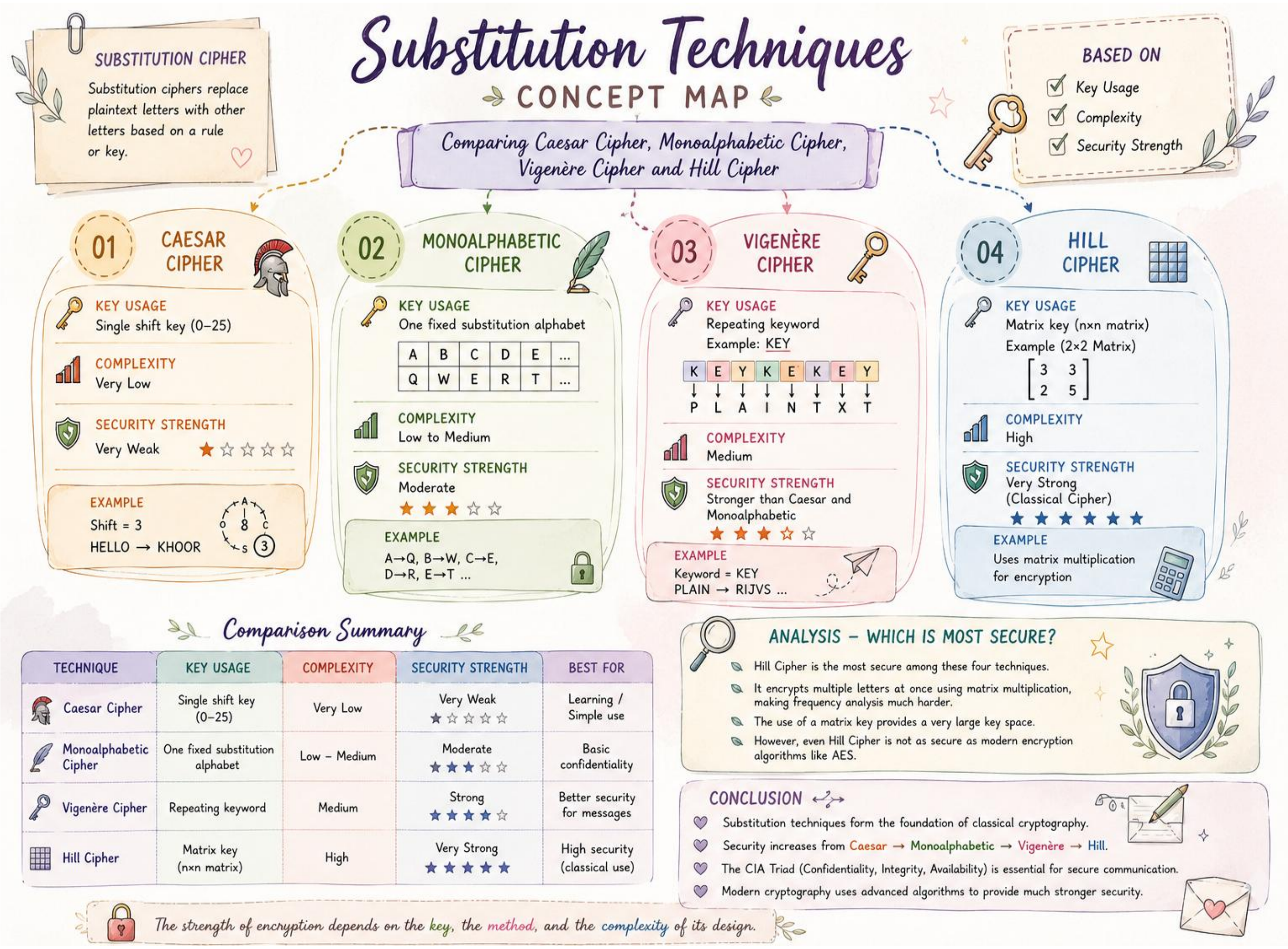
Conclusion:

1. Security attacks create threats that can lead to serious risks such as data breaches, identity theft, and service disruption.
2. The **CIA Triad** helps mitigate these risks by ensuring data confidentiality, integrity, and availability.
3. Security mechanisms such as encryption, authentication, digital signatures, firewalls, and backups strengthen system protection.
4. Understanding the relationship between attacks, threats, risks, and security goals is essential for building secure and reliable information systems.

2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.

Introduction:

1. Substitution ciphers replace plaintext characters with other characters based on a specific encryption rule.
2. Different substitution techniques vary in key usage, encryption complexity, and security strength.
3. Comparing these techniques helps understand their advantages, limitations, and practical applications.
4. Modern cryptography has evolved from these classical ciphers to provide stronger security.



Conclusion:

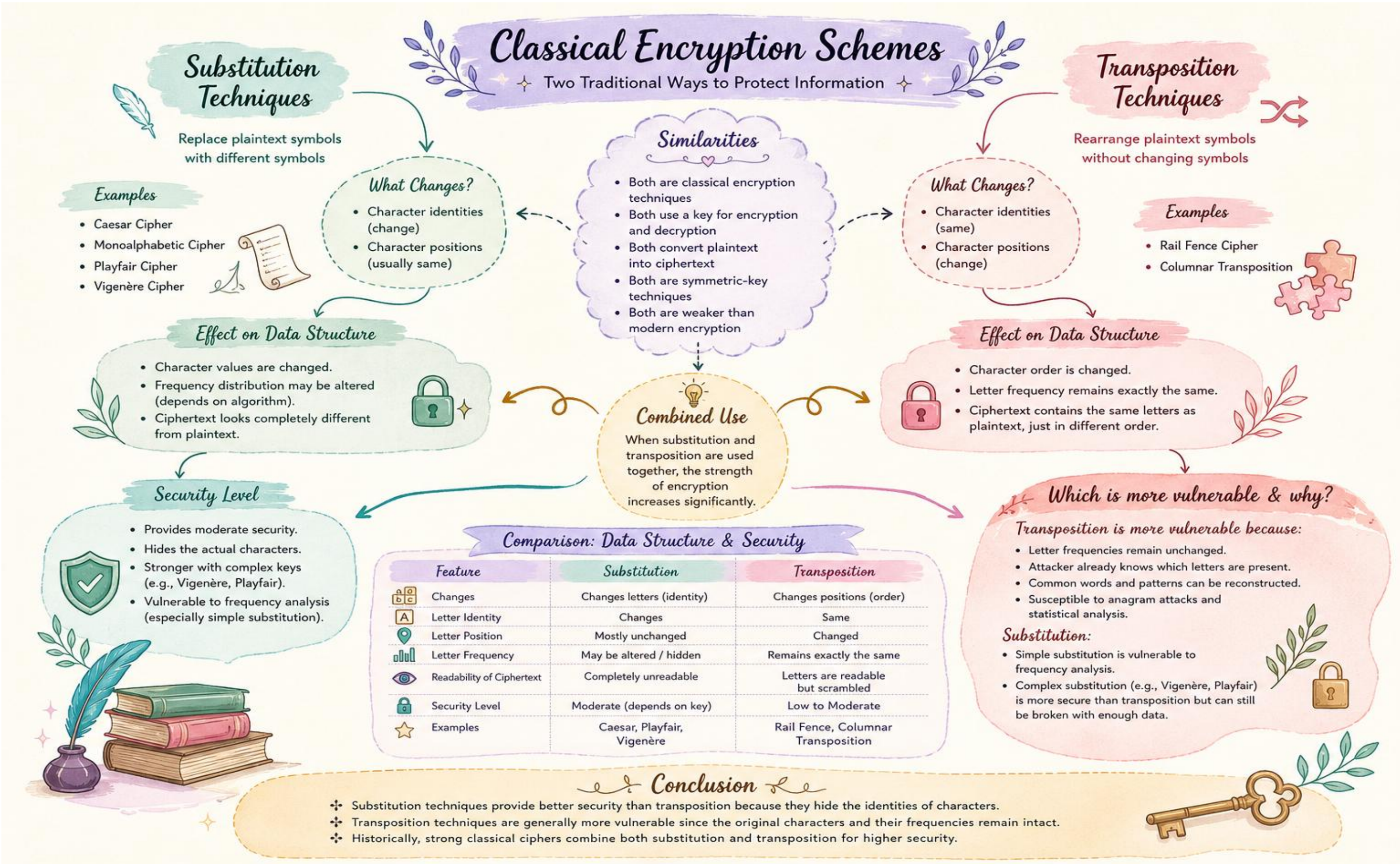
- Classical substitution techniques differ in **key usage, encryption complexity, and security strength**, making each suitable for different levels of security.
- **Caesar Cipher** is the simplest and least secure, while **Monoalphabetic** and **Vigenère Ciphers** provide improved protection against basic attacks.
- **Hill Cipher** offers the highest security among these classical techniques because it uses **matrix-based encryption**, making cryptanalysis more difficult.
- Despite their historical importance, these classical ciphers are **not sufficient for modern cybersecurity** due to advances in attack methods and computing power.
- Modern encryption algorithms such as **AES** are preferred today because they provide stronger confidentiality, better resistance to attacks, and enhanced data protection.

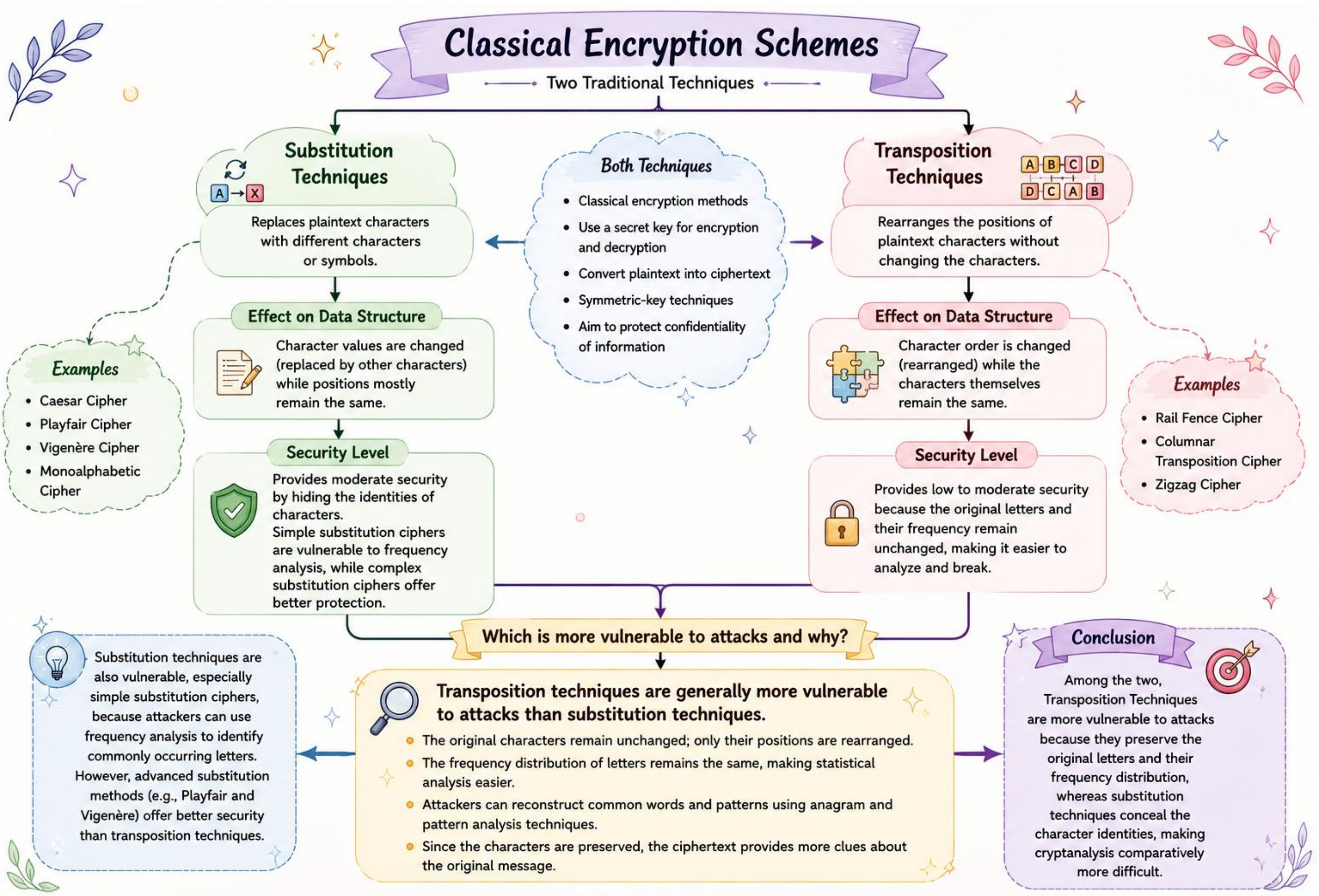
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
a. Show differences and similarities

i. Map how each affects the Data structure and Security level
ii. Analyse which is more vulnerable to attacks and why

Introduction:

- **Classical encryption schemes** are traditional cryptographic methods used to convert readable **plaintext** into unreadable **ciphertext** to ensure secure communication.
- These techniques use a **secret key** for both encryption and decryption, allowing only authorized users to access the original message.
- The two main types of classical encryption are **Substitution Techniques**, which replace characters with other characters, and **Transposition Techniques**, which rearrange the positions of characters.
- Although classical encryption methods are less secure than modern algorithms, they provide the **foundation of modern cryptography** and help in understanding the basic principles of data security.





4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:

a. Show relationships between these concepts
b. Analyze how each contributes to secure communication
c. Include one real-world application for each

Introduction:

- **Steganography** is the technique of hiding secret information within a cover medium (such as an image, audio, video, or text) so that the existence of the message remains concealed.
- It enhances **secure communication** by protecting confidential data from unauthorized access without drawing attention to the hidden information.
- The three key concepts of steganography are **Data Hiding**, **Watermarking**, and **Survivability**, each serving a unique role in ensuring security, authenticity, and robustness.
- Steganography is widely used in applications such as **secure communication, digital copyright protection, medical image authentication, military communication, and confidential data transmission**.


Steganography
The art and science of hiding information within other non-secret data.
(a) RELATIONSHIPS BETWEEN THE CONCEPTS
Data Hiding
Conceals the existence of a secret message by embedding it inside a cover medium.
Supports
Data hiding can embed watermarks for ownership tracking.
Watermarking
Embeds identifiable information (e.g., logo, ID) into digital content to prove ownership or authenticity.
Enhances
Watermarking improves survivability by ensuring the mark remains after processing or attacks.
Survivability
The ability of hidden data or watermark to remain intact and detectable after intentional or unintentional modifications.
Key Characteristics
• Focus on confidentiality
• Hides both the message and its existence
• Carrier is not altered perceptibly
Key Characteristics
• Focus on ownership & authenticity
• Watermark is usually visible (visible watermark) or invisible (invisible watermark)
• Helps in copyright protection and tamper detection
Key Characteristics
• Ensures robustness and reliability
• Withstands attacks, compression, noise, cropping, resizing, etc.
• Measures the strength of steganographic methods or watermarks
Interrelationship Summary
Data Hiding provides secrecy by concealing the message.
Watermarking adds authenticity/ownership information into the content.
Survivability ensures the hidden data or watermark remains even after changes or attacks.
(b) HOW EACH CONTRIBUTES TO SECURE COMMUNICATION
Data Hiding
Ensures covert communication by hiding the existence of the secret message inside a cover medium, so attackers are unaware that any secret information is being transmitted.
Contributes to Confidentiality
Watermarking
Ensures authenticity and integrity of the content by verifying the source and detecting tampering, which prevents unauthorized use and forgery.
Contributes to Authentication & Integrity
Survivability
Ensures reliability of the message or watermark by remaining intact under various conditions (attacks, compression, noise, etc.), allowing the information to be retrieved and trusted even after processing.
Contributes to Robustness & Trust
(c) ONE REAL-WORLD APPLICATION FOR EACH
Data Hiding
Application
Covert Military Communication
Secret messages are hidden inside images sent over public networks to avoid detection by enemies.
Watermarking
Application
Digital Media Copyright Protection
Movies, music, and images are watermarked to prove ownership and prevent illegal copying and distribution.
Survivability
Application
Medical Image Authentication
Patient details or hospital logos are watermarked in medical images to remain intact even after compression or editing.
Conclusion
Together, Data Hiding ensures secrecy, Watermarking ensures authenticity, and Survivability ensures reliability — forming a strong foundation for secure communication.

5.Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:

a. Show how each concept is mathematically connected
b. Map their role in encryption/decryption
c. Analyze which concept is most critical for public key cryptography and justify